

What We Did This Session

A simple, no-jargon walkthrough of this session's conversation - what was asked, and what happened

5 September 2026

This was the big one: building the cross-border trading module from a standing start, start to finish. It began with a scoping decision, went through real data sourcing (including a real outage on Europe's own data platform), four real versions of the model, an extension to two more years, a moment of catching our own misleading number, and two honestly-tested dead ends on the one gap that's left. Finished by updating both the Progress Report and the supervisor deck with all of it.

1. Deciding How Big to Build "The Rest of Europe"

One combined stand-in for all of Germany's neighbours, not ten separate country models.

The scoping question left over from last time: build one simplified combined zone representing everywhere Germany trades with, or ten fully separate country models. Chose the combined approach - matching how our import numbers already worked - as the right first version, with the option to split it apart later if needed.

2. Sourcing Real Data - and Riding Out a Real Outage

Pulled real demand, capacity, and weather data for Germany's real neighbours from genuine public sources.

Used the same kind of official European statistics office data already trusted elsewhere in this project to get real electricity demand and power-plant capacity figures for all 10 of Germany's real neighbouring countries. Used the same real weather-driven wind/solar service already used for Germany's own model to get real renewable output for the combined zone, properly weighted by each country's actual size.

You asked: "Retry the ENTSO-E Transparency Platform..."

Europe's official cross-border data platform genuinely went down for several hours - confirmed it was real, not a fluke, and simply waited it out.

Needed one more real number: how much power can physically flow across each border. Europe's own official platform for this started returning errors on every attempt - checked directly (including via the platform's own website, and with a freshly generated access token) to rule out a problem on our end before concluding it was a genuine platform-wide outage. Set up an automatic retry every 3 hours and worked on other real tasks (the hourly demand shape, the renewable profiles, fuel prices) while waiting, rather than sitting idle or guessing a number to keep moving.

The platform came back - then a second real snag with full-year requests, worked around cleanly.

Once it recovered, found that asking for a whole year of data in one request failed, while asking for it one month at a time worked fine - confirmed this directly by testing both side by side, then rebuilt the fetch to pull the data in monthly chunks instead. One background fetch also looked stuck for over ten minutes with no visible progress; checked the actual files being saved rather than assuming a hang, found it was quietly working the whole time, and fixed the display so future progress would be visible immediately.

In plain terms: Like a courier service's website going down right when you need to check a delivery slot - you don't cancel the delivery, you check back later and get on with other errands in the meantime.

3. The First Working Two-Way Market

Built and ran the first real version - genuinely better match quality, but a real cost too.

With enough real data in hand, built the first working version: Germany's market properly connected both ways to the new combined European zone, using a temporary placeholder for the border capacity number until the real figure was ready. Our trustworthy day-to-day match-quality score improved from 0.675 to 0.739 - a genuine gain. But the average price gap to Brainpool got worse (-11.86 to -30.39 EUR/MWh), because the other side of the border had no storage or flexibility of its own yet, and the border limit itself was still a placeholder guess rather than a real number.

4. Real Border Data Replaces the Guess

Swapped the placeholder for a real, sourced border-capacity number.

Used the real cross-border flow data (once fully recovered) to work out a genuine, grounded limit on how much power can flow between Germany and its neighbours - replacing the placeholder guess with real numbers (roughly 21,600-22,000 MW each direction). This confirmed the earlier match-quality improvement was real and not just an artefact of an oversized placeholder.

5. Adding Real Storage and Subsidies - Our Best Result Yet

Gave the other side of the border its own real storage and subsidy treatment, the same way Germany already has.

Added genuine battery-style and reservoir hydro storage to the combined European zone, sized using real published hydro-industry figures, plus real government-subsidy treatment instead of a placeholder zero. Along the way, found and fixed three genuine bugs in how the two zones were wired together - including one where two zones sharing a single forecasting agent caused the model to crash outright, and another where the wrong kind of market agent was assigned a subsidy type it couldn't actually handle. Checked directly that the new storage was doing real work, not just sitting idle.

Result: average price within 60 cents of the real figure - the best match this whole project has produced.

Average price gap improved to just -0.59 EUR/MWh (was -30.39 two versions ago), with match quality holding strong and the number of 'ran out of power' hours dropping from 26 back down to a much more reasonable 7.

In plain terms: The first version was like two countries linked by a single wire with no batteries on either end - useful, but jumpy. Adding real storage on both sides smoothed that out, the same way home batteries smooth out a household's own solar swings.

6. Testing 2028 and 2029 - and Catching a Misleading Number

Extended the same build to the two other years - match quality held up strongly in both.

Applied the exact same build to 2028 and 2029, keeping the European side's data frozen at its real 2023 shape (the same principle already proven for Germany's own out-of-sample testing) so nothing was re-tuned per year. Match quality jumped strongly in both years (2028: 0.446 to 0.718, 2029: 0.446 to 0.752), confirming the earlier win wasn't a one-year fluke.

You asked: "What is next? Also, the bias for 2029 seems worse in the coupled result."

You were right to flag it - checked properly, and it was a misleading headline number, not a real regression.

The plain average price gap for 2029 did look concerning at first glance. Broke it down properly rather than accepting the headline number: found that 88% of that gap came from just a handful of extreme 'ran out of power' hours, not from ordinary days. On ordinary days - which is over 99% of the year - all three years are excellent and remarkably consistent (2027: -0.59, 2028: +1.08, 2029: +3.21 EUR/MWh). Corrected the Progress Report in place rather than leaving the misleading version standing, since letting an all-hours-only number stand would have understated a genuinely strong result.

7. Chasing the Last Gap: Two Honest Dead Ends

The one real, remaining pattern: those rare extreme hours get more frequent the further a year is from 2027 (7 in 2027, 17 in 2028, 76 in 2029). Two reasonable explanations were tested directly, one after the other.

You asked: "Test scaling ROE/transmission to keep pace with DE's growth."

Did not work - tested directly, and it made one year worse, not better.

The idea: Germany's own demand keeps growing while the border-capacity number stays flat, so scale the border capacity up to match. Built and ran it. Result: 2028 actually got worse (17 extreme hours became 25), and 2029 barely improved (76 became 68). Dug into why: checked hour-by-hour whether Germany was actually hitting its border limit during these extreme hours, and found it wasn't - the ceiling was only actually being hit in a small minority of cases, meaning capacity was never really the true bottleneck.

You asked: "Test the MaximumShiftedEnergyPerIterationInMWH fix."

Also wrong - opened up AMIRIS's own real program code to check, and said so plainly.

The next suspect: a specific internal setting that limits how much power AMIRIS's own matching engine can shift per round. Opened the actual compiled program (not just our settings file) to check its real default value directly - and found it's already set to effectively unlimited, not a hidden cap at all. This disproved the very hypothesis raised just beforehand, and that was stated directly rather than glossed over. Found one real, unexplained clue instead while investigating: Germany's import amount during these rare hours stays suspiciously flat no matter how much spare capacity is available - pointing to the matching engine's own internal stopping logic as the real limiter. Confirming that fully would mean editing and rebuilding AMIRIS's own program, a bigger step not yet taken.

In plain terms: Two sensible theories, tested properly instead of assumed - both ruled out with real evidence, leaving one genuine clue worth chasing next, rather than an unexplained shrug.

8. Bringing It All Together

You asked: "Update the supervisor deck with everything since Phase 34. / Update the progress report PDF with Phases 34-40 too."

Both documents fully updated with this session's work.

Added nine new slides to the supervisor deck, placed right after the earlier small-idea tests and before the closing summary slides - covering the build decision, the real data sourcing story, each of the four model versions, the 2028/2029 results (the corrected, accurate version), and the two honestly-ruled-out dead ends. Regenerated the technical Progress Report the same way, which already contained the full written record of every phase from this session.

	2027	2028	2029
Match quality (with two-way trading)	0.717	0.718	0.752
Average price gap, ordinary days	-0.59	+1.08	+3.21
Rare extreme hours (goal: near zero)	7	17	76

That last row is the one honest, open thread left for the next session: a real, narrow, well-diagnosed gap, not a mystery - with a solid next lead (the matching engine's own internal stopping logic) and a clear next step (patching and rebuilding AMIRIS itself) already identified.